

Wireless Communication (EX751 / EX715)

Concise PYQ

Notes: TU IOE PYQ 2070–2082, 22 papers. Regular = **bold**, Back = normal; BEX (EX751) = normal font, BEI (EX715) = **mono**. Marks in (); unique values only. Chapter and bullet order mirrors the Detailed PYQ — every question in that document is represented here, with every distinct marks split it was ever asked for. Keys: (I)=Importance/Need (+)=Advantage (-)=Disadvantage +Fig=Figure +Eg=Example Vs=Compare/Differentiate.

1. Introduction

- Evolution of wireless / cellular 1G–5G; technology + worldwide market penetration; features introduced 2G→5G (4)(6)
- 1G/2G/3G/4G Vs, technology advancement; improvements in 2G, 3G & beyond-3G; 2G Vs 3G +Eg of technologies (with prioritized handoff & cell dragging); features of 4G over 3G (4)(5)(6)(4+2)
- Major benefits of wireless communication + the 4G system (2+4)
- Trends beyond 3G (key technological changes); forward & reverse channel; 5G Vs 4G, features of 5G (4)(2)(2+2)

2. Cellular Concepts

- Frequency reuse; co-channel reuse ratio in detail; minimizing reuse distance maximizes spectral efficiency (I); cellular concept for $N=7$ (6)(4)(3)
- Prove $Q=\sqrt{3N}$, $N=i^2+ij+j^2$; optimal N for omni / 120° / 60° sectoring at SIR 15 dB, $n=3$ or 4, + comment (4)(4+4)(3+5)
 - with co-channel Vs adjacent channel interference (3+3)
 - + 20 MHz duplex spectrum, 25 kHz simplex channel → duplex channels, channels/cell at $N=4$ (3+4+3)
 - obtain $S/I=\sqrt[3]{3N}/i_0$; min. co-channel centre distance (7-cell reuse, 23 km² cells) (4)
- Handoff — margin +Fig; handoff strategy used in GSM; handover process, types & application; practical handover considerations (3)(4)(6)(8)(4+4)
 - proper Vs improper handoff +Fig; cell dragging; two prioritized handoff schemes; define GoS, traffic intensity, average holding time, blocked call (1+3)(2)(2+4)(3+4)
- Co-channel Vs adjacent interference; cell footprint + interference tier; interference in wireless/mobile communication (2.5+2.5)(1+4)(4)
- GoS + how it is measured in a blocked-call-cleared trunking system, with optimal N (3+5)
 - 394 cells/19 ch/2% blocking → users + market penetration (5)
 - 3 calls/h of 5 min: traffic/user, users at 1% on 1 ch and on 5 trunked ch, users doubled → new blocking (Erlang-B table printed) (2+2+2+3)
 - $N=7$, 1% blocking, 57 ch, omni → 60° sector: traffic capacity loss (GOS table printed) (2+2+2)
 - 1500 km², 12 cells, 28.5 MHz → cells, channels/cell, carried traffic, users, mobiles per channel (12)
 - 32 cells, 1.6 km, reuse 7, 336 ch → area, ch/cell, simultaneous calls (4)
 - cluster size / no. of clusters / max. users — 73 Ma prints no data (6)
- Microcell zone concept (why needed + explain) (3)(1+3)
- Sectoring Vs cell splitting — which to choose as planning engineer, and why; how both improve coverage & capacity (4)(5)
- Techniques for enhancing capacity & coverage in a cellular radio network (8)

3. Radio Wave Propagation

- Free-space & link budget — received power / E-field / power flux density / rms input voltage (8)(2+2+2+2)
 - 50 W@900 MHz, 10 km, $G_T=1$, $G_R=2$, 50 Ω matched
 - 165 W@325 MHz into 12 dBi, 15 km, 6 dBi (6)
 - 10 W@250 MHz, 20 m coax @3 dB/100 m, 9 dBi, 25 km, 4 dBi, 0.2 dB mismatch (8)
 - coverage range: 150 W, threshold -104 dBm, 15 dB loss at first metre, $n=4$ (6)
 - $\lambda/4$ monopole 2.55 dB, $E=10^{-3}$ V/m@1 km, 900 MHz → length & A_e , then two-ray P_r (50 m / 1.5 m, 5 km) (6)
 - cordless 52000 kHz, 50 m → free-space path loss (6)(2+4)
 - 50 W → dBm/dBW, P_r (100 m), P_r (10 km), unity gains (1+1+2+2)
 - basic propagation models; far-field distance, 1 m antenna@900 MHz (3+3)
- Propagation mechanisms (3 basic); large- Vs small-scale model; prop-

- agation as large-scale path loss + small-scale fading; define path loss + multipath parameters used to classify fading (3)(3+6)(8)(2+4)
- Two-ray model — derive phase difference; derive path & phase difference, LOS Vs ground-reflected, by geometry; derive path loss equation +Fig with (+) over free space (8)(6)(2+6)
 - diffraction + derive phase difference in Fresnel zone geometry (2+6)
 - scattering + derive the two-ray ground-reflected model (2+8)
 - path loss Vs fading; time-dispersion fading and its types (2+6)
- Log-distance path loss model (4)
- Link budget design — BTS↔mobile reverse-link distance +diagram, Okumura, $A_{mu}=43$, $G_{area}=9$, 100 m/10 m, 10 dB margin (10)
 - feasibility of a 10 km suburban AP-client link@2.4 GHz, Okumura, $A_{mu}=20$, $G_{area}=13$ (12)
- Outdoor models — any two; Okumura & Hata necessary conditions; Okumura Vs Hata (2+2)(3+3)(4)(5)
 - Okumura: $d=50$ km, $h_{te}=100$ m, $h_{re}=10$ m, suburban, $A_{mu}=43$, $G_{area}=9$ @900 MHz (4+4)
 - same + EIRP 1 kW, $G_R=10$ dB → EIRP in dBm and P_r (4+2+2)(5)
 - Okumura: find T-R separation, $L=167$ dB@2.1 GHz, 40 m/2 m, large city, $A_{mu}=34$, $G_{area}=0$ (8)
 - Hata: medium city 950 MHz/45 m/10 km/5 m, + compare with free space (4+4)
 - Hata: large urban 900 MHz/100 m/2 m/4 km (formulae printed on the paper) (7)
 - Hata: max. cell radius & cell area, 50 W, -91 dBm, 900 MHz, 30 m/1 m, medium city (6)
 - Two-ray point-to-point 800 MHz/30 m/2 m/10 km, + compare with free space (4+4)
 - Ericsson Multiple Breakpoint model (2)
- Indoor models (any two); factors influencing the indoor model (5)(8)
- Small-scale fading — define, types, influencing factors (2+2)(4+4)(3+6)(2+4+4)
 - + derive the relation for Doppler shift (1+2+5)
 - types based on delay spread and Doppler spread (1+3)
 - parameters of the mobile multipath channel (7)
- Doppler spread + derive Doppler shift; frequency dispersion (I) + derive Doppler spread; Doppler effect in small-scale fading (2)(2+4)
- Coherence BW & coherence time — definition and significance; Doppler spread + classify fading by rms delay spread and coherence time; frequency-selective fading, coherence BW Vs delay spread (4)(2+5)(4+4)(2+7)
 - PDP (70 Ma figure): Doppler spread, fading types by Doppler spread, mean excess & rms delay spread, 90% and 50% coherence BW (4+4)
 - smallest T_s / greatest symbol rate without an equalizer ($\sigma_r/T_s \leq 0.1$); PDP (0, 0, -10, -20) dB @ (0, 50, 75, 100) μ s (8)
 - $P_r(\tau)=(-20, -10, -10, 0)$ dB @ $\tau=(0, 1, 2, 5)$ s → mean/rms/max. excess delay + 50% coherence BW (6)
 - (0, -10, -20, -23) dB @ (0, 100, 200, 400) ns → rms delay spread, 90% coherence BW; flat Vs frequency-selective for AMPS (30 kHz) & GSM (200 kHz) (6)
- Rayleigh Vs Rician fading channel +expressions; frequency-selective fading + Rayleigh distribution (2)(5)(2+4)

4. Modulation–Demodulation

- QPSK — transmission & detection +Fig, equation & constellation diagram (6)(7)(8)
- OQPSK Tx & Rx; why $\pi/4$ -QPSK is preferred over OQPSK (5+2)
- BPSK against QPSK modulation (5)
- DPSK Tx & Rx + PN sequence and why it is used (4+2+2)
- MSK as special CPFSK / OQPSK; OQPSK Vs MSK Vs GMSK (BW & power efficiency), which suits GSM and why; what is MSK modulation (2)(4)(3+3+2)
- (+) of digital over analog modulation; GMSK transmission & detection +block & constellation diagram; MSK & GMSK + OFDM modula-

- tor/demodulator +Fig (2+6)(8)
- OFDM operation +Fig (Tx and Rx); generalize modulation & demodulation; Tx alone; Rx alone; Tx +(+)(-); principle + types of spread spectrum modulation (4)(8)(4+4)
- M-ary QAM (need); 16-QAM Vs 16-PSK using constellation diagrams (2+6)
- Spread spectrum — define + DSSS +Fig; define + FHSS +Fig; DS & FH techniques (4)(6)(2+5)(2+6)
- (+) of spread spectrum; major (-) of the spread spectrum system (2)(4)

5. Equalization & Diversity

- Equalization (I); the signal processing operation that minimizes ISI (4)(2+5)(2+6)
 - + RACK receiver + structure of the linear transversal equalizer +Fig (2+1+4)
- Training & tracking modes of adaptive equalizers; why equalization is needed + both modes in detail (4)(1+3)
- MSE algorithm — optimal solution of the weights +diagram +derivation (7)(8)
- MLSE +Fig; + define time diversity and two implementations of it (4)(6+1+5)
- Adaptive equalization algorithms (any two) (4)(5)
- Interleaving — working principle +Fig; why it is needed; define + how an adaptive equalizer offsets ISI in a multipath time-dispersive channel (4)(2+6)
- Why equalization & diversity are needed; space-diversity reception methods, explain any one (3+3+2)
- Diversity (I) + types / any two in detail; max-ratio combining space diversity; combining methods for space diversity; space diversity techniques +block diagrams; how diversity improves network quality (2)(4)(2+2)(2+6)(4+4)(4+6)
- Antenna diversity techniques compared; feedback Vs maximum-ratio combining (4)
- RAKE (RACK) receiver — M-branch receiver +Fig; how it exploits time diversity (3)(4)(8)(3+5)

6. Speech & Channel Coding

- Characteristics of the speech signal; how they are used in designing coders (3)(4)(6)(8)
- Vocoder (I) + any two predictive coders; vocoders +Fig and their kinds; vocoder operation +Fig; formant vocoder + speech characteristics; why speech coding is needed + basic VOCODER concept (4)(6)(2+3)(2+6)(3+5)(4+4)
- Waveform Vs voice coding, speech characteristics, GSM CODEC +Fig (4+2+2)
- Any two source coders used in speech coding; types of frequency-domain coding of speech (4)(6)
- LPC +neat block diagram (with speech characteristics); types of LPC; sub-band coding with Tx & Rx (6)(2+6)(3+5)
- What is channel coding; 7-bit Hamming code +Eg; convolutional encoding & decoding (2)(5)
- Viterbi decoding algorithm; define turbo coding (2)(3)(5)

7. Multiple Access

- FDMA — non-linear effect + 25 MHz spectrum, 100 kHz edge guard band, 200 kHz channel → no. of channels (3+2)
 - 12.8 MHz per simplex band and 12.5 MHz total, 10 kHz guard, 30 kHz channel → no. of channels; FDMA (+)(-) (4)
- TDMA — (+) over FDMA; GSM frame efficiency (6 trailing, 8.25 guard, 26 training bits, 2 bursts of 58 data bits); N-TDMA users per cluster (25 MHz one-way, 30 kHz channels, $B_g=20$ kHz) (4)
- CDMA — characteristics, (+) & limitations; important features; design considerations; (+)(-); (+) over TDMA; merits & demerits (2)(4)(6)(2+6)
- CDMA implementation — encoding & decoding parts +Eg (7)(8)(2+6)
- Hadamard H_8 / 8×8 Hadamard (Walsh) code; steps satisfying the Hadamard condition + construct (3)(4)(3+5)
- SSMA — DS & FH techniques with (+)(-) of FDMA/TDMA/CDMA; types of SSMA + FDMA Vs CDMA; SSMA variants & application +Fig (8)(4+4)(6+2)
- FHMA — working +Eg; Tx & Rx +block diagram; operation + types of FHMA; principle + two hybrid SSMA techniques mitigating the near-far problem (4)(5)(6)(7)(4+2)(4+6)
- Hybrid SS multiple access (any two), with (+)(-); self-jamming in CDMA + FHMA +Fig + two hybrids; FHMA principle + near-far effect and its solution; define near-far + one hybrid SSMA that mitigates it; multiple access + all near-far mitigation techniques (4)(6)(2+2)(3+4)(5+4)(2+4+6)
- SDMA + two hybrid SSMA techniques minimising the near-far effect (2+6)
- Multiple access — define + TDMA/CDMA/SDMA; TCDMA and Time Division Frequency Hopping; define + FDMA Vs CDMA; FDMA Vs TDMA Vs CDMA (+)(-); TDMA Vs FDMA (4)(7)(2+4)(2+6)
- Duplexing and its different types (2+4)
- WLAN; WiFi (3)(5)

8. Wireless Systems & Standards

- GSM architecture — draw + operation of each component; components of the Network Switching Subsystem (4)(5)(8)(4+4)
- GSM traffic & control channels; BCH & CCCH; Broadcast Control Channel; operation of the different control channels (4+4)(5)(8)
- GSM frame structure + why TDMA frame efficiency cannot reach 100%; + how traffic and control channels are used in a call; frame structure; channel structure; GSM frame hierarchy (4)(5)(6)(4+4)(4+6)(6+2)
- GSM signal processing, speech ↔ radio signal; specifications of GSM (5)(8)
- CDMA IS-95 — channelization code + forward channels; draw the forward channel; draw the reverse channel; pilot & sync channels in the forward link +Fig (4)(6)
- GSM Vs CDMA standards; CDMA Vs LTE system architecture + functions of the entities (3)(4)(3+5)
- WiMAX; LTE (3)(4)(5)
- Regulatory issues in wireless communication; spectrum regulations; regulatory issues in licensing & spectrum allocation (4)(5)
- Significance of spectrum management; functions of the regulatory body (2+3)(2+4)